Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-Driven 3D Reconstruction Framework: Transforming Low-Cost Consumer Scans into Manufacturing-Ready Models for Mass Customization			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (x), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		Electrical・Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Software	Software Solutions	Measuring Instruments	Optical Measuring Instruments
Concept of Proposed Technology	The Challenge: Companies wanting to offer personalized products (custom shoes, medical devices, ergonomic equipment, etc.) face a critical problem: Getting accurate 3D scans of customers' bodies is expensive, requires expert operators, and specialized equipment. Consumer-grade scanners are affordable but produce poor-quality scans with gaps, noise, and distortions – especially when operated by untrained users. Our Solution:			

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	We have developed an intelligent software framework that transforms imperfect, low-cost 3D scans into precise, manufacturing-ready 3D models. The technology works by: 1. Smart Shape Understanding: Using AI-based statistical models that ”understand” how different body parts or objects should look, the system can intelligently fill gaps and correct errors in scans. 2. Pose & Movement Correction: When people move during scanning or hold unnatural poses, our system separates these temporary distortions from the person’s actual body shape – like distinguishing between ”bent arm during scanning” vs. ”permanently bent arm.” 3. Privacy-by-Design: Unlike cloud-based AI systems that require sending personal body data to external servers, our technology works locally and doesn’t need to store sensitive biometric information. 4. Hardware-Independent: Works with any consumer 3D scanner (smartphone apps, depth cameras, handheld scanners) – no expensive specialized equipment needed. Applications: • Medical & Orthopedics: Custom orthotics, prosthetics, compression garments • Fashion & Footwear: Personalized shoes, clothing, accessories • Sports Equipment: Custom helmets, protective gear, ergonomic equipment • Automotive & Furniture: Personalized seats, steering wheels, ergonomic workplace design • Consumer Products: Any product requiring body-fit customization
Rationale for Joint R&D Project	Why This Project Needs International Collaboration: Market Opportunity: The global mass customization market is growing rapidly, particularly in Korea’s strong manufacturing sectors (footwear, medical devices, automotive, consumer electronics). Korean SMEs are well-positioned to capitalize on this trend but need accessible technology to enable affordable customization. Technical Gap: Our research institutes have developed breakthrough algorithms for 3D reconstruction, but transforming this into a production-ready system requires: • Real-world validation across diverse manufacturing contexts • Industry-specific optimization for different product categories • Manufacturing pipeline integration knowledge • Market feedback for commercialization Why Korean SME Partnership is Essential: 1. Domain Expertise: Korean partners bring deep manufacturing knowledge and specific industry requirements 2. Test & Validation: Access to real production environments and quality standards 3. Market Intelligence: Understanding of customer needs and commercialization pathways

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	4. Co-Development: Joint refinement ensures the technology meets actual industrial needs, not just academic benchmarks
Key Words	1. 3D Shape Reconstruction & Completion 2. AI-Driven Optimization Algorithms 3. Privacy-Preserving Biometric Processing 4. Mass Customization Manufacturing 5. Low-Cost Consumer Scanner Integration

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4) – Technology validated in lab environment with proof-of concept for specific use cases	**End of the Project**	Level (7) – System prototype demonstration in operational manufacturing environment
R&D Objectives	Primary Objective: Develop a production-ready, hardware-agnostic 3D reconstruction framework that enables Korean SMEs to offer mass-customized products using affordable consumer scanning technology. Specific Technical Objectives: 1. Universal Biomechanical Shape Models • Develop AI models that can separate permanent shape characteristics from temporary poses/movements • Create models applicable across multiple body parts and object categories (not limited to single applications) • Enable automatic correction of scan artifacts caused by movement, deformation, or poor sensor quality 2. Intelligent Optimization Pipeline • Implement real-time optimization algorithms that align imperfect scan data with learned shape models • Develop adaptive algorithms that work regardless of scan quality, overlap, or completeness • Minimize computational requirements to enable deployment on standard industrial computers 3. Privacy-Preserving Architecture • Design system that processes biometric data locally without cloud dependencies • Develop anonymization techniques for model training that protect customer privacy • Ensure GDPR/PIPL compliance for international markets 4. Manufacturing Integration Framework • Create SDK/API for seamless integration into existing CAD/CAM workflows • Develop quality validation metrics that guarantee manufacturing-ready output • Enable export to standard manufacturing formats (STL, STEP, IGES, etc.) 5. Multi-Domain Validation • Validate technology across at least 3 distinct application domains • Establish quantitative quality benchmarks for different use cases • Document best practices for different scanner types and scanning conditions			
R&D Target Specifications	Key Performance Targets: • Reconstruction Accuracy: – Current State (TRL 4): ±2.0mm (controlled lab) – Target (TRL 7): ±0.5mm (production environment) • Processing Time: – Current State (TRL 4): 5-10 minutes per scan – Target (TRL 7): <60 seconds per scan • Minimum Scan Overlap: – Current State (TRL 4): 70% overlap required – Target (TRL 7): 10% overlap sufficient • Scanner Compatibility: – Current State (TRL 4): Tested with 2 devices – Target (TRL 7): Compatible with 10+ consumer devices • Operator Skill Required:			

	– Current State (TRL 4): Trained operator – Target (TRL 7): Untrained consumer/staff • Manufacturing Acceptance Rate: – Current State (TRL 4): 70% without manual cleanup – Target (TRL 7): >90% without manual cleanup System Deliverables: • Production-ready SDK/API with comprehensive documentation • Export to standard manufacturing formats (STL, STEP, OBJ, etc.) • Automated quality validation (watertight, manifold meshes) • Multi-platform support (Windows, Linux, macOS) Overall Impact Metrics: • 70% reduction in total scanning + processing time vs. professional systems • 80% reduction in equipment costs vs. professional 3D scanning solutions • 95%+ success rate processing real-world consumer scans
R&D Plan	Three-Year Development Roadmap: Year 1: Foundation & Algorithm Development (Months 1-12) Phase 1.1: Requirements & Architecture (Months 1-4) • Workshop with Korean SME partner to define requirements and use cases • Evaluate scanner hardware compatibility • Design system architecture and data protocols • Milestone: Requirements document and architecture design Phase 1.2: Core Algorithm Development (Months 5-10) • Extend existing shape models to handle biomechanical deformations • Develop pose-correction algorithms • Implement optimization pipeline for scan registration • Milestone: Working prototype demonstrating pose-independent reconstruction Phase 1.3: Initial Validation (Months 11-12) • Test algorithms with diverse scan data • Benchmark accuracy and processing time • Deliverable: Alpha version of reconstruction system Year 2: Integration & Testing (Months 13-24) Phase 2.1: SDK Development & Integration (Months 13-18) • Develop SDK/API with documentation • Implement CAD/CAM export capabilities • Test integration with partner's workflow • Milestone: Beta SDK integrated into test environment Phase 2.2: Multi-Scanner Compatibility (Months 19-21) • Test and optimize for multiple consumer scanner types • Develop calibration and quality assessment tools • Deliverable: Scanner compatibility guidelines Phase 2.3: Pilot Deployment (Months 22-24) • Deploy system in partner's production environment • Collect performance data and user feedback • Iterative improvements based on real-world use • Milestone: System validated in production setting Year 3: Optimization & Finalization (Months 25-36) Phase 3.1: Performance Enhancement (Months 25-30)

	• Optimize processing speed and accuracy • Improve robustness to scan quality variations • Enhance mesh quality for manufacturing • Milestone: Target specifications achieved Phase 3.2: Validation & Documentation (Months 31-34) • Validate across partner's application domains • Document best practices and guidelines • Benchmark against existing solutions • Deliverable: Validation report Phase 3.3: Commercialization Preparation (Months 35-36) • Finalize SDK documentation • Prepare training materials • Establish support framework • Deliverable: Production-ready SDK v1.0 Development Approach: • Iterative development with regular partner feedback • Focus on practical manufacturing requirements • Privacy-first design (local processing) • Modular architecture for flexibility and maintenance

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Institute for Optical Systems (IOS) – Prof. Dr. Georg Umlauf: Primary Focus: Geometric Modeling & 3D Reconstruction Algorithms Core Responsibilities: 1. Mathematical Foundation & Algorithms • Development of statistical shape models and biomechanical deformation frameworks • Design and implementation of surface reconstruction algorithms • Geometric optimization techniques (ICP, surface fitting, mesh processing) • Quality validation algorithms for manufacturing-ready meshes 2. 3D Geometry Processing • Point cloud registration without overlap requirements • Gap filling and noise reduction algorithms • Mesh generation and topology optimization • CAD/CAM format conversion and export 3. Research Resources Provided: • Access to computational geometry research infrastructure • Expertise in splines, subdivision surfaces, reverse engineering and reconstruction, geometry processing • Laboratory facilities for controlled testing and validation • Graduate student researchers specializing in geometry processing 4. Deliverables: • Core reconstruction engine and shape model library • Geometric processing modules • Scientific publications and documentation • Algorithm validation reports Steinbeis Transfer Center KIO – Pascal Laube: Primary Focus: AI, Optimization & Industrial Integration Core Responsibilities: 1. AI & Machine Learning Pipeline • Design and training of AI models for shape understanding • Development of adaptive optimization algorithms • Privacy-preserving machine learning techniques • Model training infrastructure and data management 2. Optimization & Performance • Real-time optimization algorithms for industrial deployment • Computational efficiency optimization • Multi-objective optimization for quality vs. speed tradeoffs • Auto-tuning and adaptive parameter optimization 3. Software Engineering & Integration • SDK/API architecture and implementation • System integration design and testing • Software quality assurance and testing frameworks • Documentation and developer tools 4. Industry Transfer & Commercialization

	• Requirements engineering with SME partners • Technology transfer methodology • Training material development • Commercial deployment support 5. Research Resources Provided: • AI/ML development infrastructure and tools • Software engineering expertise and best practices • Industrial project management experience • Access to Steinbeis transfer network 6. Deliverables: • AI/ML models and training pipelines • SDK/API implementation • Integration tools and documentation • Commercial deployment package Joint Responsibilities (IOS + KIO): • Collaborative algorithm design and validation • Joint publications and patent applications • Regular coordination meetings and progress reviews • Integrated testing and validation • Combined technical support for SME partner
Expected R&R of Korean SME Partner	Core Responsibilities: 1. Domain Expertise & Requirements Definition • Provide detailed manufacturing requirements and quality standards • Define industry-specific constraints and specifications • Share knowledge of current production workflows and pain points • Identify market opportunities and customer needs • Contribute to use case prioritization and feature requirements 2. Testing & Validation Environment • Provide access to production or pilot manufacturing facilities • Supply representative scanning hardware used in target markets • Offer real-world test scenarios and production conditions • Facilitate user acceptance testing with actual customers/operators • Provide feedback on usability and practical deployment challenges 3. Data Contribution • Contribute real-world scan data for algorithm validation (anonymized/consent-based) • Provide manufacturing quality feedback data • Share production success/failure case studies • Help establish ground-truth datasets for benchmarking 4. Manufacturing Integration • Test integration of SDK into existing CAD/CAM workflows • Validate output quality against manufacturing requirements • Provide engineering resources for integration testing • Document manufacturing-specific optimization needs • Support pilot production runs 5. Market & Commercialization Input

	• Provide market intelligence and competitive analysis • Identify additional Korean SME candidates for technology adoption • Support go-to-market strategy development • Facilitate connections with industry associations and potential customers • Provide feedback on pricing, licensing, and business model 6. Technical Collaboration • Assign dedicated engineering liaison(s) for project coordination • Participate in regular project meetings and reviews • Contribute to problem-solving and troubleshooting • Support on-site testing and validation activities • Collaborate on documentation and training materials 7. Resource Commitments • Dedicated personnel for project collaboration (engineering, production, quality) • Access to manufacturing facilities and equipment • Test materials and production resources for validation • Time commitment for joint development activities • Co-investment in project success (if applicable per TIPA guidelines) Expected Benefits for Korean SME: • Proprietary access to cutting-edge customization technology • Competitive advantage in mass customization markets • 70-80% reduction in scanning equipment costs • New revenue streams from personalized products • Enhanced brand positioning as technology innovator • Potential for technology licensing to other companies • Strengthened capabilities in AI and advanced manufacturing

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technological Achievements: 1. Novel AI-Driven 3D Reconstruction Framework • First universal biomechanical shape model system for consumer-grade scanners • Breakthrough in pose-invariant shape reconstruction • Privacy-preserving biometric processing technology • 10+ scientific publications in top-tier journals/conferences 2. Commercial Software Product • Production-ready SDK/API for mass customization • Compatible with 10+ consumer scanning devices • Integration modules for major CAD/CAM systems • Comprehensive documentation and training materials 3. Validated Industrial Process • Demonstrated 70-80% cost reduction vs. professional scanning solutions • Proven 90%+ manufacturing acceptance rate • Established best practices for multiple industries • Quality benchmarks and validation frameworks Revenue Potential: • Estimated 100+ Korean SMEs could adopt this technology

	• Average licensing value: $50K-200K per company • Total addressable Korean market: $10M+ in first 5 years • Global market potential: $100M+ within 10 years Direct Economic Benefits: • Cost Savings: 70-80% reduction in scanning equipment costs = $500K-2M saved per mid-size manufacturer • Revenue Growth: Korean SME partners can capture premium pricing for customized products (20-40% price premium) • Market Expansion: Access to new customer segments requiring customization Indirect Economic Benefits: • Supply Chain Effects: New demand for consumer scanning hardware, software services, training • Innovation Ecosystem: Strengthened Korea-Germany technology collaboration • Knowledge Transfer: Advanced AI and optimization expertise transferred to Korean industry • Patent Portfolio: Intellectual property assets for participating organizations Broader Impact: • Sustainability: Reduced waste through made-to-order production vs. mass production • Accessibility: Better medical devices for underserved populations • Industry 4.0 Advancement: Contribution to smart manufacturing and digital transformation • International Cooperation: Model for successful cross-border R&D collaboration

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization